Coal Mill (Bowl Mill) - Operation & Maintenance Manual

Vertical bowl-type pulverizer; 6 mills per 500 MW unit (5+1)

Field	Value
Document No.	MILL-OM-2210
Revision	Rev 3.1
Equipment OEM	(synthetic; BHEL class)
Applicable standard	ISO 10816 / IEC / CEA norms
Classification	O&M reference - SYNTHETIC (Ballast demo)
Owner	Plant O&M / Reliability

Disclaimer: This is a synthetic document generated for the Ballast demo. Values are grounded in public standards and typical industry practice but do not represent any specific OEM's proprietary manual.

1. Overview

Bowl mills pulverize raw coal to the fineness required for stable combustion. A 500 MW unit runs six mills in a 5+1 configuration; loss of one mill typically forces a partial derating (Criticality 'B'). Mill availability and fineness directly affect boiler efficiency and NOx.

2. Vibration & Temperature Limits

Zone	Vel. RMS (support-A)	Vel. RMS (support-B)	Action
Zone A (new/good)	<= 2.3	<= 3.5	Newly commissioned; ideal
Zone B (acceptable)	2.3 - 4.5	3.5 - 7.1	Unrestricted long-term operation
Zone C (alarm)	4.5 - 7.1	7.1 - 11.0	Short-term only; plan corrective action
Zone D (danger)	> 7.1	> 11.0	Risk of damage; trip / immediate action

Point	Normal	Alarm	Trip
Gearbox/pinion vibration	Zone A/B	4.5 mm/s	7.1 mm/s
Mill outlet temp (PA)	70 - 90 degC	95 degC	105 degC
Mill motor bearing temp	55 - 75 degC	90 degC	100 degC

3. Service Intervals

Task	Interval
Grinding roll / bull-ring inspection	4,000 running hrs
Roll & liner replacement	On wear (typ. 12,000-18,000 hrs)
Gearbox oil change	8,000 hrs
Mill motor bearing regreasing	2,000 hrs
Classifier / seal air check	Quarterly

4. Failure Modes

- Roller wear (erosion) - falling fineness, rising mill DP and vibration.
- Gearbox bearing wear (fatigue) - rising vibration at gear-mesh frequency.
- Choke / blockage - sudden mill DP rise; risk of mill trip.
- Hot primary air excursion - fire/deflagration risk; obey outlet-temp trip.